

Introduction to Statistical Learning

Assigned Reading: Read Chapters 1 and 2 of *The Elements of Statistical Learning*. These chapters give an overview of key concepts and major methods. We will delve into much greater detail on the methods, so you needn't be concerned about understanding everything now.

What is Statistical Learning?

Goals

The goal of this chapter is to lay out some basic concepts that will delineate the major methods of this course:

1. Define statistical learning.
2. Prediction versus inference.
3. Supervised and unsupervised learning.
4. Classification versus regression.
5. Parametric versus nonparametric methods.

Prediction versus Inference

This course focuses on *statistical learning*. What is statistical learning? Broadly speaking, statistical learning refers to a set of methods for learning from data. However, all data visualization, data exploration, and statistical methods have the goal of learning from data and *statistical learning* has a more precise meaning (although not necessarily a meaning on which there is universal agreement). Traditional statistical methods are typically focused on *inference*: using data to make inferences about variables or the relationships between variables. On the other hand, statistical learning is focused on *prediction*: using data to forecast unobserved outcomes or future events.

Example - breast cancer gene expression data Next, we will look at some example data to demonstrate the differences between inference and prediction.

Like most cancers, breast cancer is not homogeneous. There are different types of breast cancer that have different prognoses and require different treatment plans. Unfortunately, identifying the cancer subtype is often difficult. The data set below measures gene expression in different breast cancer tumor types. The goal is to understand how gene expression patterns change for different tumor types.

```
library(tidyverse)

bcdat=read_csv("../Data/breastcancer/Breast_GSE45827.csv")
dim(bcdat)
```

```
## [1] 151 54677
```

```
bcdat
```

```
## # A tibble: 151 x 54,677
##   samples type '1007_s_at' '1053_at' '117_at' '121_at' '1255_g_at' '1294_at'
##   <dbl> <chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1      84 basal      9.85      8.10      6.42      7.35      3.03      6.88
## 2      85 basal      9.86      8.21      7.06      7.69      3.15      7.54
## 3      87 basal     10.1      8.94      5.74      7.69      3.13      6.56
## 4      90 basal      9.76      7.36      6.48      6.99      3.18      7.80
## 5      91 basal      9.41      7.75      6.69      7.33      3.17      7.61
## 6      92 basal      7.51      8.80      6.24      7.20      2.99      7.99
## 7      93 basal     10.4      8.93      5.63      6.88      3.10      6.27
## 8      94 basal     10.2      7.81      6.70      6.92      3.14      7.52
## 9      99 basal     10.3      7.80      6.73      7.10      3.14      6.89
## 10    101 basal      9.05      8.04      7.66      6.90      2.99      7.67
## # i 141 more rows
## # i 54,669 more variables: '1316_at' <dbl>, '1320_at' <dbl>, '1405_i_at' <dbl>,
## #   '1431_at' <dbl>, '1438_at' <dbl>, '1487_at' <dbl>, '1494_f_at' <dbl>,
## #   '1552256_a_at' <dbl>, '1552257_a_at' <dbl>, '1552258_at' <dbl>,
## #   '1552261_at' <dbl>, '1552263_at' <dbl>, '1552264_a_at' <dbl>,
## #   '1552266_at' <dbl>, '1552269_at' <dbl>, '1552271_at' <dbl>,
## #   '1552272_a_at' <dbl>, '1552274_at' <dbl>, '1552275_s_at' <dbl>, ...
```

Each line of the data set corresponds to a different patient with breast cancer. The column **type** is the type of tumor. The remaining 54,000+ columns correspond to gene probes. These probes give the expression levels for the approximately 20,000 human genes. The expression level is a measure of how active the gene is in the tissue sample and is measured relative to a reference.

Breast Cancer Inferential analysis One goal that we might have with this data is to understand how gene expression patterns vary between tumor types. This could help us understand the causes of breast cancer or how the impact of breast cancer mutations leads to different tumor types. A typical analysis would look for genes whose expression varies between tumor types.

We will start by reducing the data to just the samples that are breast cancer tumor types:

```
bcdat1=filter(bcdat,type%in%c("basal","HER","luminal_A","luminal_B"))
probe_list=names(bcdat1[3:ncol(bcdat1)])
bcdat1$type=factor(bcdat1$type)
```

Next, we will use an ANOVA to test whether each gene probe is significantly different between tumor types:

```
output=data.frame(probe=probe_list,pvalue=NA)
for(i in seq_along(probe_list))
{
  probe=probe_list[i]
  output$pvalue[i]=summary(aov(bcdat1[[probe]]~type,data=bcdat1))[[1]][1,5]
}
```

It takes a while (> one hour) to run the above analysis. To avoid having to re-run repeatedly, I save the results to an external file below. Note that I have the R Markdown chunk option `eval=FALSE`. This makes it so that the code chunk is not run when I knit the document.

```
save(output,file="brca_inf_output.RData") #it takes a while to run the above analysis. To avoid having
```

I then load the results from the file below:

```
load(file="brca_inf_output.RData") #load saved object with p-values
```

Find probes that are significant under a Bonferroni correction for multiple testing:

```
signif=(output$pvalue<(0.05/length(probe_list))) #vector of TRUE and FALSE with TRUE for signif  
sum(signif)
```

```
## [1] 11120
```

There are 11,120 probes that are significantly differentially expressed between tumor types. In a typical gene expression study, we would next find the genes associated with the probes. We would then look for patterns connecting those genes and how they relate to the different tumor types and then try to use these patterns to understand something about the biology of breast cancer - how tumors are formed, what differentiates the tumor types, etc.

The statistical goal would be to identify genes that are differentially expressed between tumor types and the scientific goal would be to use the knowledge of differentially expressed genes to better understand something about cancer biology.

Breast Cancer Tumor Prediction In contrast, a different task that we might try to do with this data set is to build a prediction model for tumor type. We might do this in order to have a method to reliably identify the tumor type in order to guide treatment. Professor Schliekelman's brother formerly worked for a startup company that was doing exactly this for prostate cancer.

We will spend much of this course learning methods for building predictive models.

We will do a simple example below. First, to reduce the computational burden, I will reduce to the 1000 genes that are most significantly associated with tumor type:

```
top1000=order(output$pvalue)[1:1000] #gives indexes of probes with 1000 lowest -values  
bc1000=bcdat1[,c(2,top1000+2)] #column number for the probe is index+2
```

Next, I will randomly divide the data into training and test data. We will use the training data to fit the model and the test data to test how well the model predicts the tumor type:

```
train=sample(c(TRUE,FALSE),nrow(bcdat1),replace = TRUE)  
test=!train  
  
bc1000_train=bc1000[train,]  
bc1000_test=bc1000[test,]
```

I will use the K-nearest neighbors method for predicting the breast cancer tumor type. We will discuss the K-nearest neighbors method in detail later in the semester. Briefly, this method classifies the tumor type for a new sample by finding the K nearest neighbors in the training data to the sample in the space of the

predictor variables. It classifies the new sample as the class that is most common among the K nearest neighbors.

Let's do this in R:

```
library(class)

knn.pred1=knn(train=bc1000_train[,-1],test=bc1000_test[,-1],cl=bc1000_train$type,k=10)
```

The table below compares the actual to predicted tumor types for the test data

```
table(bc1000_test$type,knn.pred1)
```

##		knn.pred1			
##		basal	HER	luminal_A	luminal_B
##	basal	17	0	0	0
##	HER	2	14	0	0
##	luminal_A	0	0	10	0
##	luminal_B	0	0	0	12

We see that in this example the prediction is highly accurate. Note how this is different from the above inferential analysis. In the above analysis, our goal is to draw conclusions about individual genes. Here, we don't care about individual genes. Instead, we are using the genes collectively to make predictions about tumor type.

Supervised versus Unsupervised learning

Statistical learning is most often *supervised learning*. In a supervised learning setup, we have a set of predictor variables X and a response variable Y that are related in some unknown way. We have

$$Y = f(X) + \epsilon$$

We are then typically interested in finding out about $f(X)$, in either an inferential sense or a predictive one. In contrast, in *unsupervised learning*, there is no response variable Y . In this case, the goal is to learn about relationships between samples or between predictors without a response variable to guide the way.

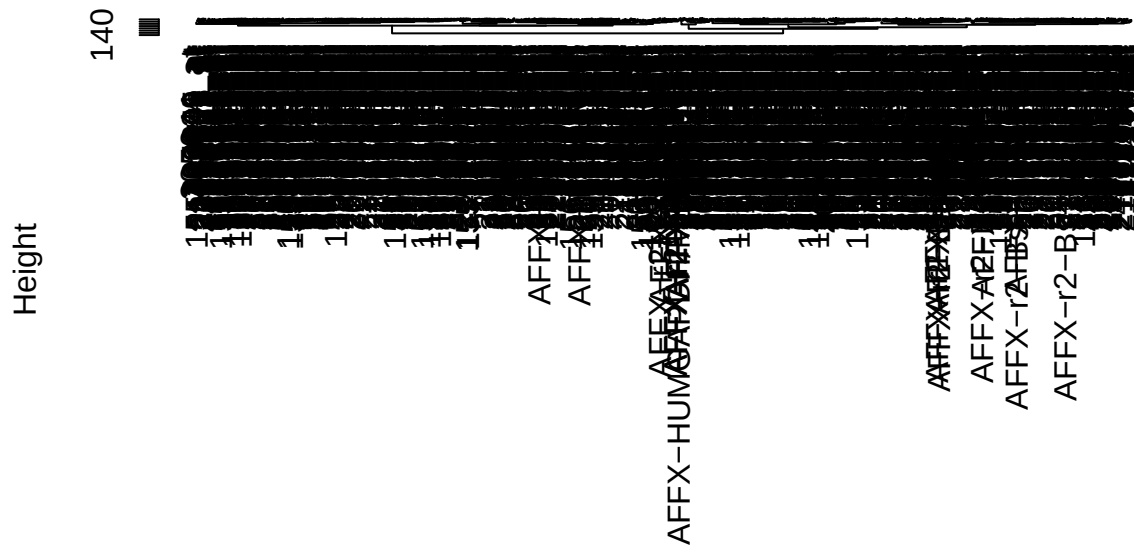
The most common type of unsupervised analysis is cluster analysis. We can use cluster analysis both to look for relationships among samples and for relationships among predictor variables. In the case of gene expression data, we can use cluster analysis to 1) look for relationships among tumors in terms of shared patterns of expression across genes, or 2) look for relationships among genes in terms of shared patterns of expression across tumors.

We will discuss clustering methods later in the semester. In the analysis below, I use hierarchical clustering, which is one of the simplest clustering methods. A Euclidean distance is calculated between each pair of genes in the gene expression space using the `dist` function. Genes are clustered in a hierarchical fashion according to the pairwise distances using the `hclust` function. At each stage, the closest pair of objects are clustered.

```
hclustBC=hclust(dist(t(bc1000[,-1])))

plot(hclustBC)
```

Cluster Dendrogram



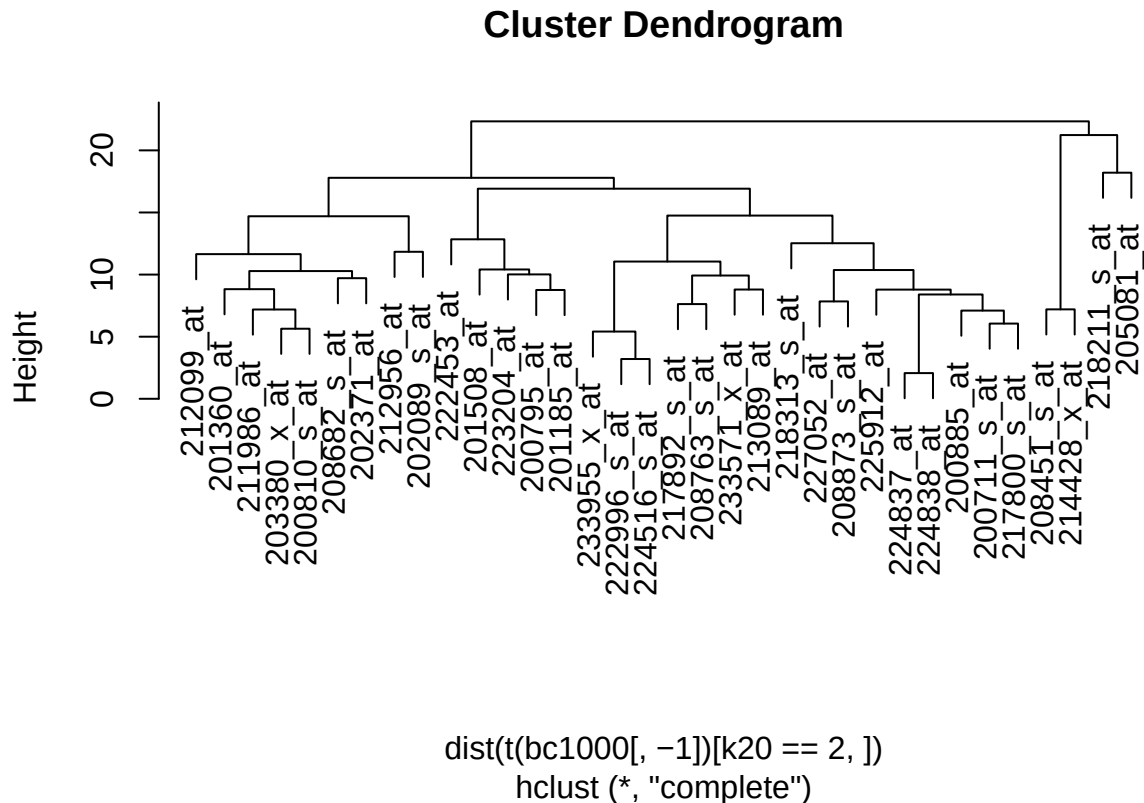
```
dist(t(bc1000[, -1]))
hclust (*, "complete")
```

This results in a plot that is hard to read because of the large number of genes. We will zoom into one part of the cluster tree in order to see better.

```
k20=cutree(hclustBC,k=20)

hclustBC_k2=hclust(dist(t(bc1000[, -1])[k20==2,]))

plot(hclustBC_k2)
```



This shows the relationship between a set of probes. Cluster analysis is commonly used with gene expression data. It is known that genes work together in groups and cluster analysis helps identify those groups.

Regression Versus Classification

Another key distinction is between regression and classification. We use regression methods when the response is continuous and classification when the response is categorical.

The above example with breast cancer tumor prediction is an example of **classification**. We are using a set of quantitative variables (gene expression) in order to predict a categorical variable (tumor type). In general, **classification** involves the use of a set of features (continuous or categorical variables) in order to predict the value of a categorical variable.

In contrast, we use regression methods when the response is continuous. Everyone in this course is familiar with regression and so I won't waste time with an example.

Parametric versus nonparametric methods

A final key distinction is between parametric and nonparametric methods. *Parametric* methods are based on an explicit model for the relationship between predictor and response. Linear regression is the most well known example. In linear regression, we assume that there is a linear relationship between the predictor variables and response, with normally distributed noise. In this course we will consider both different models for the relationship between predictors and response (i.e. nonlinear relationships) and different possibilities for the distribution of the random component (i.e. non-normal).

Nonparametric methods do not assume a specific model. The K-nearest neighbors method used above is an example of such a method. It does not assume any model for the relationship between predictors and response. We will learn many such approaches during this course.

Course Topics

This course is focused primarily on prediction, but will include some topics on inference as well. The major topics of the course are listed below, along with how they fit into the above scheme.

1. Monte Carlo Simulation
2. Model Assessment and Validation
3. Nonlinear Methods (supervised regression, parametric and nonparametric)
4. Dimension Reduction Techniques (supervised regression, parametric and nonparametric)
5. Classification Methods (supervised classification, parametric and nonparametric)
6. Clustering Methods (unsupervised, parametric and nonparametric)
7. Ensemble Methods (supervised, regression or classification, parametric and nonparametric)